2024, Mar. 2024 Release LOC:Acute Adult
Labor and Delivery

Utilization Benchmarks: *Multiple options available, see table below for options*

Overview

Instruction: This subset covers routine labor and delivery (including home birth that requires admission within 24 hours), labor induction, and delivery due to actual or potential maternal or fetal compromise of a fetus greater than or equal to 14 weeks gestation (including the delivery of a stillborn fetus), and complications during the immediate postpartum period delaying discharge which include:

- Bleeding
- Eclampsia
- Fever
- Hemolysis, elevated liver enzymes, and low platelet count (HELLP) syndrome
- Postpartum hypertension
- Preeclampsia
- Post-dural puncture headache
- Wound dehiscence

If the patient develops a condition necessitating the need for hospitalization not covered within this subset, refer to the appropriate subset and conduct a review. For example, if the patient develops pneumonia, refer to the Infection: Pneumonia subset.

Evaluation/Treatment:
The intrapartum period involves close monitoring of labor progression, maternal stability, and fetal status. Treatment may include pain management, maintaining hydration, or interventions to ensure labor and delivery progress without fetal or maternal compromise. The rate of postpartum recovery depends on the delivery technique and complications occurring during the antepartum, intrapartum, or postpartum period. Routine maternal evaluation includes surveillance of vital signs, fluid status, blood loss, contraction of the uterus, pain, and signs of infection.

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:

- ⊖ **Episode Day 1, One:** [1]
  (Symptom or finding within 24h)
  - ⊖ **ACUTE, One:** [2]
    - ⊖ Labor, **One:** [3]
      - ⊖ Induction of labor, ≥ **One:** [4]
        - − Mechanical methods [5, 6]
        - − Pharmaceutical methods (includes PO, buccal, intracervical, intravaginal, sublingual) [7, 8, 9]
      - ⊖ Prelabor rupture of membranes (PROM) confirmed and, **All:** [10, 11, 12, 13]
        - − ≥ 37 wks gestation
        - − Fetal heart rate monitoring [14]
        - − Uterine monitoring [15]
      - ⊖ Term labor and, ≥ **One:** [16]
        - − Contractions ≤ every 5 min for ≥ 30 sec
        - − Cervical dilation ≥ 4 cm
        - − Cervical effacement ≥ 50%(0.50)
    - ⊖ Delivery, **One:** [3]
      - − Cesarean section
      - − Vaginal
  - ⊖ **CRITICAL, ≥ One:** [17]
    - − Antihypertensive, continuous [18, 19]
    - − DIC and blood product transfusion [20]
    - − Eclampsia and neurological assessment every 1-2h [21, 22]
    - − ECMO or ECLS [23]
    - ⊖ Heart failure and, **Both:**
      - ⊖ Finding, ≥ **One:**
        - − Arterial Po_2 < 56 mmHg(7.4 kPa)
        - − O_2 sat < 89%(0.89) and < baseline [24]
      - − Diuretic every 1-2h or continuous
    - ⊖ Hemodynamic instability, **All:** [25]
      - ⊖ ED treatment, ≥ **One:**
        - − ≥ 2 units blood product transfusion
        - − ≥ 2 liters IV fluid [26]
        - − ≥ 1 unit blood product transfusion and ≥ 1 liter IV fluid
      - ⊖ Findings after ED treatment, ≥ **One:**
        - − Heart rate > 120/min
        - − Systolic blood pressure < 100 mmHg and < baseline
        - − MAP < 65 mmHg
      - − Requiring ongoing resuscitation
    - − Hemodynamic monitoring, invasive [27, 28]
    - ⊖ HELLP syndrome, **Both:** [29]
      - ⊖ Finding, ≥ **One:**
        - ⊖ Lab values, **All:**
          - − Platelet count < 50,000/cu.mm($50x10^9$/L)
          - − AST or ALT > 2x ULN
          - − LDH ≥ 600 IU/L(10.02 µkat/L)
        - ⊖ Increased risk of bleeding, ≥ **One:**
          - − PT ≥ 1.5x ULN or INR ≥ 2.0
          - − PTT ≥ 1.5x ULN
      - ⊖ Intervention, ≥ **One:**
        - − Blood product transfusion
        - ⊖ Laboratory monitoring, **All:**

- CBC
- LFTs
- Platelet count
- HFNC [30, 31, 32]
- Mechanical ventilation
- NIPPV [33]
- Oxygen ≥ 40%(0.40) [30]
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

- **Episode Day 2, One:** [1]
  - **ACUTE,** ≥ **One:**
    - Labor, **One:** [34]
      - Induction of labor, ≥ **One:** [4]
        - Mechanical methods [5, 6]
        - Pharmaceutical methods (includes PO, buccal, intracervical, intravaginal, sublingual) [7, 8, 9]
      - Term labor with contractions and progressive cervical changes ≤ 2d [35]
    - Delivery, **One:** [34]
      - Cesarean section ≤ 4d
      - Vaginal ≤ 2d
    - Post Critical care ≤ 24h
  - **INTERMEDIATE,** ≥ **One:** [36]
    - Eclampsia and neurological assessment every 3-4h [21, 22]
    - HFNC and respiratory monitoring every 3-4h [31, 32]
    - NIPPV [33]
  - **CRITICAL,** ≥ **One:**
    - Antihypertensive, continuous [18, 19]
    - DIC and blood product transfusion [20]
    - Eclampsia and neurological assessment every 1-2h [21, 22]
    - ECMO or ECLS [23]
    - Heart failure and, **Both:**
      - Finding, ≥ **One:**
        - Arterial Po_2 < 56 mmHg(7.4 kPa)
        - O_2 sat < 89%(0.89) and < baseline [24]
      - Diuretic every 1-2h or continuous
    - Hemodynamic instability persisting after ≥ 1L of IVF resuscitation or ≥ 1 unit blood product transfusion and, **Both:** [25]
      - Finding, ≥ **One:**
        - Heart rate > 120/min
        - Systolic blood pressure < 100 mmHg and < baseline
        - MAP < 65 mmHg
      - Intervention, ≥ **One:**
        - Blood product transfusion
        - IV fluid resuscitation ≥ 1L ≤ 24h [26]
    - Hemodynamic monitoring, invasive [27, 28]
    - HELLP syndrome, **Both:** [29]
      - Finding, ≥ **One:**
        - Lab values, **All:**
          - Platelet count < 50,000/cu.mm($50x10^9$/L)
          - AST or ALT > 2x ULN
          - LDH ≥ 600 IU/L(10.02 µkat/L)
        - Increased risk of bleeding, ≥ **One:**
          - PT ≥ 1.5x ULN or INR ≥ 2.0
          - PTT ≥ 1.5x ULN
      - Intervention, ≥ **One:**

- Blood product transfusion
- ⊖ Laboratory monitoring, **All:**
  - CBC
  - LFTs
  - Platelet count
- HFNC and respiratory monitoring every 1-2h [30, 31, 32]
- Mechanical ventilation
- ⊖ NIPPV and, ≥ **One:** [33]
  - FiO_2 > 50%(0.50)
  - Respiratory intervention every 1-2h [37]
- Oxygen ≥ 40%(0.40) [30]
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

⊖ **Episode Day 3, One:** [1]
- ⊖ **ACUTE, One:**
  - ⊖ **Responder**, discharge expected today if clinically stable last 12h, **All:** [38]
    - Afebrile
    - Post delivery (vaginal delivery or cesarean section)
    - ⊖ Complication or comorbidity, **One:**
      - No complication or active comorbidity relevant to this episode of care [39]
      - ⊖ Complication or comorbidity and clinically stable for discharge, ≥ **One:**
        - Bleeding controlled or resolved
        - ⊖ HELLP syndrome and, **Both:**
          - Blood pressure controlled last 24h and within acceptable limits [40]
          - Lab values within acceptable limits [40]
        - Hypertension controlled last 24h and within acceptable limits [40]
        - Preeclampsia or postpartum hypertension resolved
        - Post-dural puncture headache controlled
        - Wound with adequate healing and established wound care regimen
  - ⊖ **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
    - ⊖ Labor, **One:** [34]
      - ⊖ Induction of labor, ≥ **One:** [4]
        - Mechanical methods [5, 6]
        - Pharmaceutical methods (includes PO, buccal, intracervical, intravaginal, sublingual) [7, 8, 9]
      - Term labor with contractions and progressive cervical changes ≤ 2d
    - ⊖ Delivery, **One:** [34]
      - Cesarean section ≤ 4d
      - Vaginal ≤ 2d
    - ⊖ Vaginal bleeding > 1 pad/h and, ≥ **One:**
      - Hct or Hb monitoring
      - IV fluid or blood product transfusion
      - Ultrasound assessment [41]
    - ⊖ Fever and, **Both:**
      - Temperature ≥ 100.4°F(38.0°C) [42]
      - ⊖ Intervention, **One:**
        - Known source and anti-infective
        - ⊖ New onset and, ≥ **One:** [43]
          - Culture pending ≤ 2d
          - Imaging study ≤ 24h
    - ⊖ HELLP syndrome, **All:** [29]
      - AST or ALT > 2x ULN
      - Platelet count 50,000-100,000/cu.mm(50-100x10^9/L)
      - LDH ≥ 600 IU/L(10.02 µkat/L)
      - ⊖ Blood pressure management, ≥ **One:** [44, 45]

- Antihypertensive (includes PO) [46]
- Blood pressure monitoring at least 3x/24h
- ⊖ Laboratory monitoring, **All:**
  - CBC
  - LFTs
  - Platelet count
- ⊖ Postpartum hypertension or preeclampsia and, **Both:** [45, 47, 48, 49, 50]
  - ⊖ Finding, ≥ **One:**
    - Systolic blood pressure ≥ 140 mmHg or diastolic blood pressure ≥ 90 mmHg ≥ 2 measurements [45]
    - AST or ALT ≥ 2x ULN
    - ⊖ Creatinine, ≥ **One:**
      - > 1.1 mg/dL(97.2 μmol/L)
      - > 2x baseline [24]
    - Headache, new onset, severe, persistent, and unresponsive to medication [51]
    - Platelet count < 100,000/cu.mm(100x10^9/L)
    - Visual disturbances, persistent [52]
  - ⊖ Intervention, ≥ **One:**
    - Antihypertensive (includes PO) [46]
    - Magnesium sulfate [53]
    - ⊖ Monitoring, **Both:**
      - Blood pressure monitoring at least 3x/24h [45]
      - ⊖ Laboratory monitoring, **All:**
        - CBC
        - Creatinine
        - LFTs
        - Platelet count
- ⊖ Post-dural puncture headache and, ≥ **One:** [54]
  - Epidural blood patch performed within last 24h [55]
  - Conservative management ≤ 24h [56, 57]
  - Pharmacological management [58, 59]
- ⊖ Wound dehiscence and, ≥ **One:** [60]
  - Repair performed within last 2d
  - Wound care and wound management teaching ≤ 24h
- Post Critical care ≤ 24h

⊖ **INTERMEDIATE, ≥ One:**
- Eclampsia and neurological assessment every 3-4h [21, 22]
- HFNC and respiratory monitoring every 3-4h [31, 32]
- NIPPV [33]

⊖ **CRITICAL, ≥ One:**
- Antihypertensive, continuous [18, 19]
- DIC and blood product transfusion [20]
- Eclampsia and neurological assessment every 1-2h [21, 22]
- ECMO or ECLS [23]
- ⊖ Heart failure and, **Both:**
  - ⊖ Finding, ≥ **One:**
    - Arterial Po_2 < 56 mmHg(7.4 kPa)
    - O_2 sat < 89%(0.89) and < baseline [24]
  - Diuretic every 1-2h or continuous
- ⊖ Hemodynamic instability persisting after ≥ 1L of IVF resuscitation or ≥ 1 unit blood product transfusion and, **Both:** [25]
  - ⊖ Finding, ≥ **One:**
    - Heart rate > 120/min
    - Systolic blood pressure < 100 mmHg and < baseline

- MAP < 65 mmHg
- ⊖ Intervention, ≥ **One:**
  - Blood product transfusion
  - IV fluid resuscitation ≥ 1L ≤ 24h [26]
- Hemodynamic monitoring, invasive [27, 28]
- ⊖ HELLP syndrome, **Both:** [29]
  - ⊖ Finding, ≥ **One:**
    - ⊖ Lab values, **All:**
      - Platelet count < 50,000/cu.mm($50x10^9$/L)
      - AST or ALT > 2x ULN
      - LDH ≥ 600 IU/L(10.02 μkat/L)
    - ⊖ Increased risk of bleeding, ≥ **One:**
      - PT ≥ 1.5x ULN or INR ≥ 2.0
      - PTT ≥ 1.5x ULN
  - ⊖ Intervention, ≥ **One:**
    - Blood product transfusion
    - ⊖ Laboratory monitoring, **All:**
      - CBC
      - LFTs
      - Platelet count
- HFNC and respiratory monitoring every 1-2h [30, 31, 32]
- Mechanical ventilation
- ⊖ NIPPV and, ≥ **One:** [33]
  - FiO_2 > 50%(0.50)
  - Respiratory intervention every 1-2h [37]
- Oxygen ≥ 40%(0.40) [30]
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

⊖ **Episode Day 4-5, One:** [1]
- ⊖ **ACUTE, One:**
  - ⊖ **Responder**, discharge expected today if clinically stable last 12h, **All:** [38]
    - Afebrile
    - Post delivery (vaginal delivery or cesarean section)
    - ⊖ Complication or comorbidity, **One:**
      - No complication or active comorbidity relevant to this episode of care [39]
      - ⊖ Complication or comorbidity and clinically stable for discharge, ≥ **One:**
        - Bleeding controlled or resolved
        - ⊖ HELLP syndrome and, **Both:**
          - Blood pressure controlled last 24h and within acceptable limits [40]
          - Lab values within acceptable limits [40]
        - Hypertension controlled last 24h and within acceptable limits [40]
        - Preeclampsia or postpartum hypertension resolved
        - Post-dural puncture headache controlled
        - Wound with adequate healing and established wound care regimen
  - ⊖ **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
    - ⊖ Delivery, **One:** [34]
      - Cesarean section ≤ 4d
      - Vaginal ≤ 2d
    - ⊖ Vaginal bleeding > 1 pad/h and, ≥ **One:**
      - Hct or Hb monitoring
      - IV fluid or blood product transfusion
      - Ultrasound assessment [41]
    - ⊖ Fever and, **Both:**
      - Temperature ≥ 100.4°F(38.0°C) [42]

- Intervention, **One:**
  - Known source and anti-infective
  - New onset and, ≥ **One:** [43]
    - Culture pending ≤ 2d
    - Imaging study ≤ 24h
- HELLP syndrome and, **All:** [29]
  - Platelet count 50,000-100,000/cu.mm(50-100x10^9/L)
  - Blood pressure management, ≥ **One:** [44, 45]
    - Antihypertensive (includes PO) [46]
    - Blood pressure monitoring at least 3x/24h
  - Laboratory monitoring, **All:**
    - CBC
    - LFTs
    - Platelet count
- Postpartum hypertension or preeclampsia and, ≥ **One:** [45, 47, 48, 49, 50]
  - Antihypertensive (includes PO) [46]
  - Magnesium sulfate [53]
  - Blood pressure monitoring at least 3x/24h [45]
- Post-dural puncture headache and, ≥ **One:** [54]
  - Epidural blood patch performed within last 24h [55]
  - Conservative management ≤ 24h [56, 57]
  - Pharmacological management [58, 59]
- Wound dehiscence and, ≥ **One:** [60]
  - Repair performed within last 2d
  - Wound care and wound management teaching ≤ 24h
- Post Critical care ≤ 24h

- **INTERMEDIATE, ≥ One:**
  - Eclampsia and neurological assessment every 3-4h [21, 22]
  - HFNC and respiratory monitoring every 3-4h [31, 32]
  - NIPPV [33]
- **CRITICAL, ≥ One:**
  - Antihypertensive, continuous [18, 19]
  - DIC and blood product transfusion [20]
  - Eclampsia and neurological assessment every 1-2h [21, 22]
  - ECMO or ECLS [23]
  - Heart failure and, **Both:**
    - Finding, ≥ **One:**
      - Arterial Po_2 < 56 mmHg(7.4 kPa)
      - O_2 sat < 89%(0.89) and < baseline [24]
    - Diuretic every 1-2h or continuous
  - Hemodynamic instability persisting after ≥ 1L of IVF resuscitation or ≥ 1 unit blood product transfusion and, **Both:** [25]
    - Finding, ≥ **One:**
      - Heart rate > 120/min
      - Systolic blood pressure < 100 mmHg and < baseline
      - MAP < 65 mmHg
    - Intervention, ≥ **One:**
      - Blood product transfusion
      - IV fluid resuscitation ≥ 1L ≤ 24h [26]
  - Hemodynamic monitoring, invasive [27, 28]
  - HELLP syndrome, **Both:** [61]
    - Finding, ≥ **One:**
      - Lab values, **All:**

- Platelet count < 50,000/cu.mm(50x10^9/L)
- AST or ALT > 2x ULN
- LDH ≥ 600 IU/L(10.02 µkat/L)
- Increased risk of bleeding, ≥ **One:**
  - PT ≥ 1.5x ULN or INR ≥ 2.0
  - PTT ≥ 1.5x ULN
- Intervention, ≥ **One:**
  - Blood product transfusion
  - Laboratory monitoring, **All:**
    - CBC
    - LFTs
    - Platelet count
- HFNC and respiratory monitoring every 1-2h [30, 31, 32]
- Mechanical ventilation
- NIPPV and, ≥ **One:** [33]
  - FiO_2 > 50%(0.50)
  - Respiratory intervention every 1-2h [37]
- Oxygen ≥ 40%(0.40) [30]
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

- **Episode Day 6, One:**
  - **ACUTE, One:**
    - **Responder**, discharge expected today if clinically stable last 12h, **All:** [38]
      - Afebrile
      - Post delivery (vaginal delivery or cesarean section)
      - Complication or comorbidity, **One:**
        - No complication or active comorbidity relevant to this episode of care [39]
        - Complication or comorbidity and clinically stable for discharge, ≥ **One:**
          - Bleeding controlled or resolved
          - HELLP syndrome and, **Both:**
            - Blood pressure controlled last 24h and within acceptable limits [40]
            - Lab values within acceptable limits [40]
          - Hypertension controlled last 24h and within acceptable limits [40]
          - Preeclampsia or postpartum hypertension resolved
          - Post-dural puncture headache controlled
          - Wound with adequate healing and established wound care regimen
    - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**
      - **Use Extended Stay subset**
      - Refer for secondary review
  - **INTERMEDIATE, One:**
    - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**
      - **Use Extended Stay subset**
      - Refer for secondary review
  - **CRITICAL, One:**
    - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**
      - **Use Extended Stay subset**
      - Refer for secondary review

Discharge, Level of Care, One: [62]

- **HOME, Both:**
  - Level of care appropriateness, **Both:**

    - Home environment safe and accessible [63]
    - Patient or caregiver demonstrates ability to manage care
  - Discharge planning, **All:**
    - Provide comprehensive written discharge and teaching instructions [64]
    - Medication reconciliation [65]
    - Follow-up care planned [66]
    - Identify and address transportation needs
- **HOME CARE, Both:**
  - Level of care appropriateness, **All:**
    - Home environment safe and accessible [63]
    - Patient or caregiver willing and able to learn care
    - Outpatient management contraindicated or unavailable [67, 68]
    - Skilled services, ≥ **One:** [69]
      - Skilled nursing [70]
      - Skilled therapy, PT, OT, or ST
      - Behavioral health [71]
  - Discharge planning, **All:**
    - Provide comprehensive written discharge and teaching instructions [64]
    - Medication reconciliation [65]
    - Follow-up care planned [66]
    - Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both:**
  - Level of care appropriateness, **One:**
    - Support systems available [72]
  - Skilled treatment, ≥ **One:**
    - Clinical assessment [73]
    - Individual, group, or family counseling
    - Psychiatric, substance, or medication evaluation [74]
- **SKILLED MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
    - Treatment precluded in a lower level
    - Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 1-2x/24h [70]
      - Therapy, **All:**
        - Goal directed therapy with rehabilitation potential [75, 76]
        - Functional impairments requiring at least supervision [77]
        - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [65]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
    - Treatment precluded in a lower level
    - Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 4h/24h [70]
      - Therapy, **All:**
        - Goal directed therapy with rehabilitation potential [75, 76]
        - ≥ 2 functional impairments requiring at least minimum assistance [77]
        - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline

- Complete prior to facility transfer, **All:**
  - Medication reconciliation [65]
  - Obtain and complete forms for facility
  - Obtain discharge summary and transmit to facility and medical practitioner
  - Arrange transportation
- **SUBACUTE REHAB, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Treatment precluded in a lower level
    - Services, **All:**
      - Goal directed therapy with rehabilitation potential [75, 76]
      - ≥ 2 functional impairments requiring at least minimum assistance [77]
      - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [65]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **ACUTE REHAB, Both:**
  - Level of care appropriateness, **Both:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Services, **All:**
      - Goal directed therapy with rehabilitation potential [75, 76]
      - ≥ 2 functional impairments requiring at least minimum assistance [77]
      - Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines [78]
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [65]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
  - Level of care appropriateness, **Both:**
    - Treatment precluded in a lower level
    - In lieu of acute or continued hospitalization or failed lower level of care, ≥ **One:**
      - Primary condition or illness and treatment of active comorbid condition(s) [79]
      - Ventilator dependent ≥ 6h/d and weaning planned [80, 81]
      - Acute and comorbid conditions requiring prolonged hospitalization [82]
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [65]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation

Utilization Benchmarks

Condition or Procedure	LOS (days)	Type
776 POSTPARTUM AND POST ABORTION DIAGNOSES WITHOUT O.R. PROCEDURES	**2.4**	**CMS GMLOS**
779 ABORTION WITHOUT D&C	**1.8**	**CMS GMLOS**

Condition or Procedure	LOS (days)	Type
783 CESAREAN SECTION WITH STERILIZATION WITH MCC	4.5	CMS GMLOS
784 CESAREAN SECTION WITH STERILIZATION WITH CC	3.1	CMS GMLOS
785 CESAREAN SECTION WITH STERILIZATION WITHOUT CC/MCC	2.5	CMS GMLOS
786 CESAREAN SECTION WITHOUT STERILIZATION WITH MCC	4.3	CMS GMLOS
787 CESAREAN SECTION WITHOUT STERILIZATION WITH CC	3.3	CMS GMLOS
788 CESAREAN SECTION WITHOUT STERILIZATION WITHOUT CC/MCC	2.7	CMS GMLOS
805 VAGINAL DELIVERY WITHOUT STERILIZATION OR D&C WITH MCC	2.8	CMS GMLOS
806 VAGINAL DELIVERY WITHOUT STERILIZATION OR D&C WITH CC	2.3	CMS GMLOS
807 VAGINAL DELIVERY WITHOUT STERILIZATION OR D&C WITHOUT CC/MCC	2	CMS GMLOS
831 OTHER ANTEPARTUM DIAGNOSES WITHOUT O.R. PROCEDURES WITH MCC	3.3	CMS GMLOS
832 OTHER ANTEPARTUM DIAGNOSES WITHOUT O.R. PROCEDURES WITH CC	2.5	CMS GMLOS
833 OTHER ANTEPARTUM DIAGNOSES WITHOUT O.R. PROCEDURES WITHOUT CC/MCC	1.9	CMS GMLOS
HELLP SYNDROME (HEMOLYSIS, ELEVATED LIVER ENZYMES, LOW PLATELETS)	4.3	InterQual

Notes:

1:

Care Facilitation:

- Ensure the patient's symptoms, concerns, and needs are heard and include the patient as part of the care team (Behm et al., J Womens Health (Larchmt) 2022, 31: 1677-85)
- Complete cardiac, hypertension, preeclampsia, obstetric hemorrhage, sepsis/shock, and substance abuse risk assessment and manage patient according to protocols (Chambers et al., J Cardiovasc Dev Dis 2022, 9:; McCutcheson-Adams, Cardiac Conditions in Obstetric Care Change Package. 2022)
- Ensure patient receives respectful and equitable care (Sufrin, ACOG Committee Opinion: Importance of Social Determinants of Health and Cultural Awareness in the Delivery of Reproductive Health. Reaffirmed 2021)
    - Provide access to interpreter services
    - Assess patient's health literacy
- Educate patients on urgent warning signs and access to immediate medical care when applicable, including (Behm et al., J Womens Health (Larchmt) 2022, 31: 1677-85; Friedman et al., AJP Rep 2018, 8: e79-e84):
    - Headache that will not go away or worsens over time
    - Dizziness or fainting
    - Thoughts about hurting yourself or your baby
    - Changes in vision
    - Fever
    - Trouble breathing
    - Chest pain or fast heart rate
    - Severe and persistent belly pain
    - Severe nausea and vomiting
    - Baby's movements stop or slow down
    - Vaginal bleeding or fluid leaking
    - Swelling, redness, or pain in leg(s)
    - Extreme swelling of hands or face
    - Overwhelming tiredness
- Assess patient for chronic health conditions (e.g., hypertension, diabetes, depression, congenital heart disease, cardiovascular risk) and manage appropriately (Chambers et al., J Cardiovasc Dev Dis 2022, 9:; McCutcheson-Adams, Cardiac Conditions in Obstetric Care Change Package. 2022)
- Screen all pregnant and postpartum people for substance abuse disorders and refer to community services and resources
- Discharge with postpartum care plan and use communication hand-off tools including (Obstet Gynecol 2018, 131: e140-e50):
    - Contact information for care team
    - Follow-up appointment time, date, location, and phone number (best practice is phone or telehealth appointment 3 weeks postpartum with comprehensive visit 6-12 weeks postpartum)
    - Infant feeding plan with contact to community support (e.g., Women, Infants and Children (WIC), lactation services, mother support groups)
    - Reproductive plan (e.g., access to contraception)
    - Education on potential complications and postpartum problems
    - Treatment plan for chronic health conditions and contact information for specialist care team
    - Anticipatory guidance on signs and symptoms of perinatal depression or anxiety and management of previous mental health issues
    - Referral to specialist care, mental health support, and substance use disorder treatment if needed
- Screen for SDOH and refer to social services and/or community resources (Billoux, NAM Perspectives 2017:)
    - **Social Determinants of Health (SDOH)** are conditions in a person's environment that may affect their health. Social needs related to SDOH may impact the decision to admit, care coordination during hospital stay, as well as discharge destination. The following social needs should be assessed on admission to determine which interventions are relevant to the

individual and their family support system. This list may not be all inclusive, and all relevant SDOH factors should be considered. Any social need determined to be relevant to a patient and family should be planned for throughout the hospital stay. The following interventions should be considered for all patients and families if social need is relevant.

- **Alcohol or substance use:**
    - Coordinate care with outpatient primary care or behavioral health providers
    - Refer to addiction social support systems
    - Provide information about community resources and programs for family-centered treatment (e.g., intensive case management, detox, outpatient, residential treatment, day treatment, and medication-assisted treatment)
    - Provide information about home visitation programs (e.g., Parent-Child Assistance Program (PCAP) for pregnant and parenting individuals with substance use disorders after care) (Hildebrandt et al., Infant Ment Health J 2020, 41: 677-96; Vanderplasschen et al., Front Psychiatry 2019, 10: 186)
    - Provide information about cognitive behavioral therapy (CBT), multidimensional family therapy (MDFT), and/or pharmacotherapy for youth with co-occurring emotional/mental disorders (Substance Abuse and Mental Health Administration, Treatment Considerations for Youth and Young Adults with Serious Emotional Disturbances and Serious Mental Illnesses and Co-occurring Substance Use. 2021)

- **Child maltreatment, abuse, or intimate partner abuse, sexual, and/or psychological abuse:**
    - Arrange for alternative housing situation, foster care, or shelter if home environment is unsafe
    - Assist with creating a long-term safety plan or make appropriate referral
    - Refer to child protective services (CPS) and other domestic violence services if warranted
    - Refer for psychosocial support if warranted
    - Provide community resources and legal information
    - Provide information on parent training programs such as the Nurse-Family Partnership (NFP) (Nurse-Family Partnership National Resources for Families. 2021; Gubbels et al., Int J Environ Res Public Health 2019, 16:)

- **End of life planning:**
    - Arrange palliative care or hospice
    - Consider palliative care for children with end-stage disease
    - Consider hospice care for terminally ill children with a life expectancy of less than 6 months

- **Environmental exposure to toxic substances and other physical hazards:**
    - Lead poisoning: Provide community resources for primary prevention and secondary prevention strategies (e.g., blood lead level testing, follow-up) (Centers for Disease Control and Prevention, Lead Poisoning Prevention. 2019)
    - Tobacco use (e.g., cigarettes, e-cigarettes, vaping): Provide education and/or brief counseling about tobacco use cessation and the risk of second-hand smoke (U.S. Department of Health and Human Services, Tobacco Use in Children and Adolescents: Primary Care Interventions. 2021; Opinion, Tobacco and Nicotine Cessation During Pregnancy. 2017)

- **Financial resources strained or limited:**
    - *Food insecurity*
        - Provide information about local food banks and special nutritional programs (e.g., Special Supplemental Nutrition Program for Women, Infants, and Children (WIC), U.S. Department of Agriculture Summer Food Service Program) (Testa and Jackson, Ann Epidemiol 2021, 58: 22-8; Opinion, Tobacco and Nicotine Cessation During Pregnancy. 2017)
        - Refer to direct income assistance benefits and indirect income assistance programs (e.g., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)

- *Income insecurity*
  - Provide information about local food banks and special nutritional programs (e.g., Special Supplemental Nutrition Program for Women, Infants, and Children (WIC), U.S. Department of Agriculture Summer Food Service Program) (Testa and Jackson, Ann Epidemiol 2021, 58: 22-8; Opinion, Tobacco and Nicotine Cessation During Pregnancy. 2017)
  - Refer to direct income assistance benefits and indirect income assistance programs (e.g., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)

- **Infants and children with special health care needs:**
  - Provide anticipatory guidance and education to caregivers regarding medical condition and expected progress after discharge over the next few days, months, or years, as warranted (Dosman and Andrews, Paediatr Child Health 2012, 17: 75-80)
  - Refer for follow-up care to neonatologist or other medical subspecialist for infants or children with ongoing and/or chronic medical issues (e.g., bronchopulmonary dysplasia, feeding dysfunction) (Pediatrics 2008; 122(5): 1119-1126)
  - Refer for adaptive/assistive medical equipment and equipment for technology-dependent children (e.g., home oxygen, ventilators, monitors, tube feeding) (Pediatrics 2008; 122(5): 1119-1126)
  - Refer for neurodevelopmental assessment for high-risk infants
  - Refer to post-discharge follow-up programs (e.g., Maternal, Infant, and Early Childhood Home Visiting Program, Early Head Start program for infants or toddlers under the age of 3 and pregnant individuals in need of medical, mental health, nutrition, and education, or the Nurse-Family Partnership for mothers pregnant with their first child and for ongoing nurse care visits through the child's second birthday) (Health Resources & Services Administration, Maternal and Child, The Maternal, Infant, and Early Childhood Home Visiting Program 2021; Nurse-Family Partnership National Resources for Families. 2021)
  - Refer for behavioral support and/or counseling for caregivers through therapy groups, telephone support, and/or peer support groups for caregivers of children with similar health needs (Reyes et al., Health Equity 2021, 5: 100-18; Substance Abuse and Mental Health Administration, Treatment Considerations for Youth and Young Adults with Serious Emotional Disturbances and Serious Mental Illnesses and Co-occurring Substance Use. 2021; Pierron et al., BMC Public Health 2018, 18: 1087)
  - Refer to lactation consultant in the hospital and home visits for infants at risk for poor lactation or suboptimal infant breast-feeding behavior (Leeman et al., Pediatr Qual Saf 2019, 4: e130)
  - Refer for respite services, if warranted

- **Mental health comorbidities**:
  - Arrange appointment with primary care provider or behavioral health specialist upon discharge
  - Coordinate care with outpatient behavioral health providers
  - Provide information and resources about suicide prevention, including the 988 Suicide and Crisis Lifeline
  - Refer for psychiatric services for suicide risk screening and intervention
  - Refer to school-based mental health interventions, if applicable (Fazel et al., Lancet Psychiatry 2014, 1: 377-87)
  - Provide information about intensive community-based treatment (e.g., functional family therapy, intensive child and adolescent and psychiatric services, multidimensional family therapy, intensive case management)
  - Transfer to day treatment program, residential addiction services, or partial hospitalization when warranted

- **Risk for treatment nonadherence**:
  - Provide information about educational and behavioral support initiatives for children/individuals at risk of medication nonadherence and their caregivers, such

as text message reminders to take medication and electronic medication dispensers
    - Provide information about peer support groups for children, individuals, and/or families with poor insight into illness or noncompliance with treatment (Mental Health America, Peer Support: Research and Reports 2021; National Institute for Health and Care Excellence (NICE). Psychosis and schizophrenia in adults: prevention and management. CG178. London: NICE; 2014.)
    - Refer to interventions to prevent medical relapse or readmission, and to enhance medication compliance

- **Risk for readmission:**
    - Consider transfer to an alternative level of care such as subacute care
    - Consider home care services
    - Consider early childhood home visiting programs (e.g., Early Head Start; Early On; Nurse-Family Partnership; and the Maternal, Infant, and Early Childhood Home Visiting Program) (Health Resources & Services Administration, Maternal and Child, The Maternal, Infant, and Early Childhood Home Visiting Program 2021; Nurse-Family Partnership National Resources for Families. 2021)
    - Consider telehealth and web-based training programs (e.g., telepsychiatry or telebehavioral health and mobile health interventions) (American Academy of Child and Adolescent Psychiatry, J Am Acad Child Adolesc Psychiatry 2017, 56: 875-93)
    - Consider intensive in-home behavioral health services
    - Provide patient and family with condition-specific information, including appropriate planning for prevention of respiratory syncytial virus (RSV) infection and appropriate immunizations

- **Special health risk behaviors in adolescence and young adulthood**:
    - Arrange follow-up appointment with primary care provider and/or specialists upon discharge
    - Coordinate care with outpatient behavioral health providers
    - Provide information on mentoring programs (e.g., Big Brothers Big Sisters of America)
    - Refer for intensive community-based treatment for individuals with comorbid behavioral health issues
    - Refer to community resources for pregnancy prevention/family planning (e.g., use of birth control, subsequent pregnancies) (Maness et al., J Adolesc Health 2016, 58: 636-43)

- **Eating disorders:**
    - Provide information on nutrition services, including education and resources for diet management, physical activity, and promotion of healthy eating attitudes and behaviors (Gato-Moreno et al., Int J Environ Res Public Health 2021, 18:; Al-Khudairy et al., Cochrane Database Syst Rev 2017, 6: Cd012691)
    - Provide information on outpatient psychosocial interventions and family-based treatment (Lock and La Via, J Am Acad Child Adolesc Psychiatry 2015, 54: 412-25)

- **Parenting resources:**
    - Provide information on family resource centers and parenting support groups for young, first-time parents (e.g., National Parent Helpline) (National Parent and Youth Helpline)
    - Provide information on community resources to improve parenting skills (U.S. Department of Health and Human Services, Parenting Resources. 2021)

- **Unstable living environment or homelessness expected upon discharge:**
    - Assist with health insurance application
    - Provide information about homeless shelter application if applicable
    - Refer to local housing coordinator or case manager for immediate link to permanent supportive housing and coordinated access system (Pottie et al., CMAJ 2020, 192: E240-E54)
    - Refer to federally sponsored outpatient programs

- Refer to intensive case management intervention, assertive community treatment, or other intensive community treatment (Ponka et al., PLoS One 2020, 15: e0230896; Vijverberg et al., BMC Psychiatry 2017, 17: 284)

2:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

3:

Instruction: Labor and delivery criteria may be applied if the delivery of a stillborn or nonviable fetus is warranted. These criteria may also be applied to a home delivery that requires admission within 24 hours.

4:

In the United States, cervical ripening is usually performed as an inpatient procedure (Saad et al., Obstet Gynecol 2022, 140: 584-90). Limited information is available on the safety of outpatient management of induction of labor. Existing research acknowledges its limitations and that further extensive, larger studies and analyses are necessary to corroborate findings and establish effective and safe protocols (Sanchez-Ramos et al., American Journal of Obstetrics and Gynecology 2023; Pierce-Williams et al., Obstet Gynecol 2022, 139: 255-68; Saad et al., Obstet Gynecol 2022, 140: 584-90; McDonagh et al., Obstet Gynecol 2021, 137: 1091-101; Obstet Gynecol 2009, 114: 386-97).

5:

Mechanical induction of labor methods may include one or more of the following:

- Amniotomy
- Osmotic/hygroscopic dilators (e.g., Dilapan-S, laminaria)
- Single- or double-balloon catheters
- Extra-amniotic saline infusion

6:

Mechanical methods for induction of labor include osmotic/hygroscopic cervical dilators (e.g., Dilapan-S, laminaria), single- or double-balloon catheters with or without extra-amniotic saline infusion, and amniotomy (Sanchez-Ramos et al., American Journal of Obstetrics and Gynecology 2023; Obstet Gynecol 2009, 114: 386-97). The primary means of cervical ripening with mechanical methods is through physical forces which dilate the cervix. A Foley catheter with a French (Fr) gauge between 14 and 26 Fr with balloon inflation volumes of 30 to 80 mL and a double-balloon device may be utilized (Obstet Gynecol 2009, 114: 386-97). A Cook catheter may also be used (Sanchez-Ramos et al., American Journal of Obstetrics and Gynecology 2023). Extra-amniotic saline infusion can be added to the catheter to increase the separation of the decidua from the uterine wall. Laminaria is a naturally occurring sea plant that slowly absorbs moisture and expands when placed in the cervix, causing dilation. Synthetic osmotic/hygroscopic cervical dilators slowly absorb moisture and can expand for hours. The sequential use of a pharmacological agent commonly follows mechanical dilators for cervical ripening. However, the concomitant use of mechanical and pharmacological methods has been explored. Amniotomy, or artificial rupture of membranes, refers to rupturing the membranes using a plastic hook and can be used alone or in conjunction with cervical ripening or as a part of labor induction for individuals with a favorable cervix. Amniotomy stimulates oxytocin release from the pituitary gland and the release of prostaglandins locally (Sanchez-Ramos et al., American Journal of Obstetrics and Gynecology 2023).

7:

Pharmaceutical induction of labor methods may include one or more of the following:

- Prostaglandins (e.g., misoprostol, dinoprostone)
- Oxytocin
- Mifepristone

8:

Pharmaceutical induction of labor methods may include the use of synthetic prostaglandins E_1 and E_2 (e.g., misoprostol, dinoprostone) and oxytocin (Sanchez-Ramos et al., American Journal of Obstetrics and Gynecology 2023; Obstet Gynecol 2009, 114: 386-97). Literature notes that mifepristone exhibits efficacy in facilitating cervical ripening and can serve as a supplementary treatment alongside nonpharmacologic or pharmacologic methods for cervical ripening in individuals with an unfavorable cervix (Sanchez-Ramos et al., American Journal of Obstetrics and Gynecology 2023).

9:

Apart from dilatation and evacuation, induction of labor is another option for delivering the stillborn fetus. The appropriate labor induction techniques vary depending on the gestational age of the fetus at the time of its demise. It is recommended that healthcare practitioners examine the risks and benefits of each method in a given clinical circumstance, as well as relevant institutional experience and resources. In cases of fetal demise, shared decision making is crucial in choosing the best delivery strategy. Regardless of the cervical Bishop score, vaginal misoprostol appears to be the most effective induction technique before 28 weeks of gestation; however, high-dose oxytocin infusion is also a suitable option. When compared to misoprostol alone, there is high-quality evidence to support the use of mifepristone plus misoprostol for managing pregnancy loss before 20 weeks gestation (Obstet Gynecol 2020, 135: e110-e32).

10:

There is insufficient evidence to justify the routine use of prophylactic antibiotics with prelabor rupture of membranes (PROM) at term (i.e., greater than or equal to 37 weeks gestation) in the absence of an indication for group B streptococci (GBS) prophylaxis. It is recommended that pregnant individuals at term with PROM who test positive for GBS receive antibiotics promptly for GBS prophylaxis without an unnecessary delay to await the onset of labor (Obstet Gynecol 2020, 135: e80-e97).

11:

Prelabor rupture of membranes (PROM) refers to the rupture of membranes prior to the onset of labor, specifically occurring at term, which is defined as a gestational age of 37 weeks or greater (Obstet Gynecol 2020, 135: e80-e97).

12:

Diagnostic testing to confirm rupture of membranes includes evaluation of one or more of the following:

- Fern (+)
- Nitrazine (+)
- Amniotic fluid protein immunoassay (+)
- Pooling of amniotic fluid
- Intra-amniotic dye (+)

13:

The diagnosis of membrane rupture is typically confirmed by a clinical examination involving the observation of amniotic fluid accumulating in the vagina, a vaginal fluid pH test with nitrazine paper, and/or the identification of ferning in dried vaginal fluid upon microscopic examination. Additional tests, such as immunoassays or transabdominal intra-amniotic dye instillation, can aid in the diagnosis of prelabor rupture of membranes and preterm prelabor rupture of membranes in equivocal cases (Obstet Gynecol 2020, 135: e80-e97).

14:

The fetal heart rate (FHR) fluctuates constantly in response to changes in the intrauterine environment and other stimuli (e.g., uterine contractions). Surveillance of FHR and FHR patterns is used to assess fetal well-being and oxygenation to identify potential uteroplacental and fetal compromise, allowing for intervention, prevention, and mitigation of adverse outcomes (American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2021, 137: e116-e27; Devane et al., Cochrane Database Syst Rev 2017, 1: CD005122). Standard FHR monitoring techniques include auscultation, Doppler ultrasound, and electronic fetal monitoring that captures the FHR on a graphical printout called a cardiotocograph using an ultrasound transducer externally attached to the maternal abdomen or an electrode placed directly on the fetal scalp internally (Devane et al., Cochrane Database Syst Rev 2017, 1: CD005122). FHR monitoring is a component of the nonstress test, contraction stress test, biophysical profile, and modified biophysical profile (American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2021, 137: e116-e27).

15:

Tocography is used to detect uterine contractions and is most commonly performed externally, although internal monitoring may be indicated in select cases. External tocography, which records uterine activity, involves the placement of a transducer over the uterine fundus.

16:

Medical literature lacks sufficient evidence, has conflicting findings, or presents ambiguous results regarding the frequency and duration of contractions and the degree of cervical dilation and effacement that signify labor at term and warrants evaluation and management at the Acute level of care. InterQual® external peer reviewers agree that the majority of pregnant individuals with contractions less than or equal to 5 minutes apart for greater than or equal to 30 seconds, cervical dilation greater than or equal to 4 cm, and/or cervical effacement greater than or equal to 50%(0.50) are appropriate for further assessment and intervention at the Acute level of care. These criteria reflect current best practices and the opinion of InterQual's® external peer reviewers and are the result of an iterative process involving multiple clinicians with diverse expertise in a variety of practice settings and geographic locations.

17:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

18:

Emergent consultation with an anesthesiologist, maternal-fetal medicine subspecialist, or critical care subspecialist to discuss further interventions is recommended when first-line agents are unsuccessful in controlling acute-onset severe hypertension (Obstet Gynecol 2019, 133: e26-e50; Committee on Obstetric, Obstet Gynecol 2017, 129: e90-e5). Second-line agents for refractory management of acute severe hypertension include a continuous infusion of labetalol, nicardipine, esmolol, or sodium nitroprusside. Sodium nitroprusside should be reserved for extreme emergencies and for a short duration due to cyanide or thiocyanate toxicity risk (Griffin et al., Clin Chest Med 2022, 43: 471-88; Committee on Obstetric, Obstet Gynecol 2017, 129: e90-e5).

19:

Instruction: Medication administration includes continuous infusion and titration of medications. For medications that are titrated, the rate and dose are adjusted based on clinical monitoring and laboratory results.

20:

Disseminated intravascular coagulation or DIC is a hematological emergency where there is abrupt systemic cascade activation of coagulation leading to thrombosis of midsize to small blood vessels rapidly leading to acute organ dysfunction. Simultaneously, there is consumption of platelets and proteins involved in the coagulation cascade, resulting in acute thrombocytopenia and low clotting factors. DIC can be caused by conditions such as sepsis, trauma, severe obstetrical complications, and malignancies. Laboratory abnormalities found in DIC include low fibrinogen (not always present), elevated fibrin degradation products, elevated D-dimer, thrombocytopenia, and prolonged prothrombin (PT) and activated partial thromboplastin times. Prompt therapy for DIC includes treatment against the underlying disease but may also include platelet transfusion, plasma (fresh frozen plasma; FFP) or factor transfusion in cases of hemorrhage, or anticoagulants in situations with thrombosis formation predominates (Wada et al., J Thromb Haemost 2013:).

21:

Eclampsia is the convulsive manifestation of the hypertensive disorders of pregnancy, is among the more severe expressions of the disease, and is a significant cause of maternal mortality as seizures can lead to severe maternal hypoxia, trauma, and aspiration pneumonia. New-onset tonic-clonic, focal, or multifocal seizures in the absence of other etiologies (e.g., epilepsy, cerebral arterial ischemia) characterize eclampsia. Initial treatment for eclampsia involves preventing injury and aspiration, positioning the patient in a lateral decubitus position, administering oxygen, as well as close clinical monitoring. Magnesium sulfate is recommended as a first-line agent if it is not contraindicated (e.g., myasthenia gravis, hypocalcemia, cardiac ischemia) for the reduction of eclampsia and prevention of recurrent convulsions, as it has been shown to be more efficacious than phenytoin, diazepam, and nimodipine and is associated with reduced maternal and neonatal morbidity (Obstet Gynecol 2020, 135: e237-e60). Once maternal stabilization is achieved, expeditious delivery is recommended to reduce maternal-fetal and maternal-neonatal morbidity (Obstet Gynecol 2020, 135: e237-e60; Obstet Gynecol 2019, 133: e26-e50).

22:

A neurological assessment establishes a baseline so that subtle changes can be monitored. A comprehensive neurological assessment often includes an evaluation of:

- Mental status (e.g., level of consciousness, orientation, insight, calculation ability)
- Cranial nerves (e.g., olfactory, optic, vestibulocochlear)
- Motor system (e.g., muscle tone, strength, reflexes)
- Sensory system (e.g., light touch, pain, temperature)
- Coordination (e.g., orchestration and fluidity of movement)
- Gait (e.g., heel-to-toe straight-line walking)

23:

Extracorporeal membrane oxygenation (ECMO) or extracorporeal life support (ECLS) provides oxygen and removes carbon dioxide from blood. ECMO/ECLS facilitates the recovery of cardiopulmonary function when other measures fail to improve gas exchange. Indications for the use of ECMO/ECLS in children include refractory septic or cardiogenic shock, cardiac arrest as a bridge to transplant, or refractory hypoxemic or hypercarbic respiratory failure. There is clear evidence to support the use of ECMO/ECLS in the pediatric population, but definitive studies in adults are lacking. However, in adults, observational studies and other data support the use of ECMO/ECLS for severe refractory, hypoxemic, or hypercarbic respiratory failure (Mehta et al., American College of Cardiology Perspectives and Analysis 2018; Napp et al., Herz 2017, 42: 27-44; Guirand et al., J Trauma Acute Care Surg 2014, 76: 1275-81). ECMO/ECLS incorporates the use of an external circuit, a cannula for vascular access, a pump to propel the blood through the circuit, and an artificial membrane that oxygenates the blood. Common complications include hemorrhage, hemolysis, and infection. Stroke, embolism, and seizures in children have also been reported. Infants with intraventricular hemorrhage (IVH) or those at high risk for IVH (i.e., weight less than 2000 grams, age less than 35 weeks) are not candidates for this therapy (Abrams et al., J Am Coll Cardiol 2014, 63: 2769-78; Domico et al., Pediatr Crit Care Med 2012, 13: 16-21; MacLaren et al., Intensive Care Med 2012, 38: 210-20; Peek et al., Health Technol Assess 2010, 14: 1-46; Mugford et al., Cochrane Database Syst Rev 2008: CD001340).

24:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

25:

The mainstay of treatment of patients with hemorrhagic shock is blood product transfusion (e.g., packed red blood cells). Despite the general principle of minimizing the use of intravenous fluids in patients with known hemorrhage, crystalloids are often administered until blood products are available, especially in patients who are hypotensive (e.g., mean arterial pressure less than 65). Vital sign changes may underestimate blood loss in patients. Patients with a 30%(0.30) loss of total blood volume may have little to no change in systolic blood pressure (SBP) and only demonstrate modest tachycardia (i.e., heart rate greater than 100). By the time an SBP has dropped to less than 100 mmHg, there may be as much as a 40%(0.40) loss of total blood volume.

26:

Fluid resuscitation involves the administration of repeated intravenous fluid boluses to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as Lactated Ringer's (Myburgh, N Engl J Med 2018, 378: 862-3).
Bolus volumes for administration may be generally defined by the following:

- Fluid boluses of 250 to 1,000 milliliters in adults (Scales, Br J Nurs 2014, 23; Myburgh and Mythen, N Engl J Med 2013, 369: 1243-51; Vincent and De Backer, N Engl J Med 2013, 369: 1726-34)
- Fluid boluses of 10 to 20 milliliters per kilogram in pediatric patients. In some cases, boluses of 60 milliliters per hour or more may have to be administered in the first hour (Davis et al., Crit Care Med 2017, 45: 1061-93)

For patients with decreased cardiac reserve, significant renal dysfunction, or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children: Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020). The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., J Trauma Acute Care Surg 2014, 76: 457-61; Holodinsky et al., Crit Care 2013, 17: R249; Kirkpatrick et al., Intensive Care Med 2013, 39: 1190-206). Dynamic measures such as bedside vena cava ultrasound and monitoring of urine output may assist in determining when volume resuscitation has been achieved (Kent et al., J Surg Res 2013, 184: 561-6; Zengin et al., Am J Emerg Med 2013, 31: 763-7; Weekes et al., Acad Emerg Med 2011, 18: 912-21).

27:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

28:

The routine use of invasive hemodynamic monitoring in individuals whose sex at birth is female with severe preeclampsia or eclampsia is not recommended. However, in patients with severe cardiac or renal involvement, hypertension unresponsive to therapy, or pulmonary edema, invasive hemodynamic monitoring may be useful in guiding care (American College of Obstetricians and Gynecologists. Obstet Gynecol 2013, 122: 1122-31).

29:

Hemolysis, elevated liver enzymes, and low platelet count (HELLP) syndrome is a severe form of preeclampsia associated with increased maternal morbidity and mortality. The following criteria are diagnostic benchmarks used by many clinicians to make the diagnosis: lactate dehydrogenase (LDH) 600 IU/L(10.02 μkat/L) or more, aspartate aminotransferase (AST), and alanine aminotransferase (ALT) elevated more than twice the upper limit of normal, and platelet count less than 100,000/cu.mm($100x10^9$/L) (Louis et al., American Journal of Obstetrics and Gynecology 2022, 227: B2-B24; Obstet Gynecol 2020, 135: e237-e60). HELLP syndrome is frequently characterized by progressive and sometimes abrupt maternal deterioration (Obstet Gynecol 2020, 135: e237-e60). The management of HELLP syndrome requires very close monitoring, the availability of obstetric critical care, and personnel with specialized knowledge. Various hematologic and hepatic changes can occur in individuals with severe preeclampsia or HELLP syndrome. Platelet activation, aggregation, and consumption can lead to thrombocytopenia and represent a disease severity marker. Elevated serum concentrations of LDH can be a sign of hemolysis, and elevated LDH, AST, and ALT serum levels indicate hepatic dysfunction. Evaluation of coagulation parameters is useful when the platelet count is below 150,000/cu.mm($150x10^9$/L) with significant liver dysfunction. Abnormalities of prothrombin time, partial prothrombin time, and fibrinogen are reflective of hepatic synthetic dysfunction and advanced preeclampsia

(Obstet Gynecol 2020, 135: e237-e60).

30:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO_2) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO_2 with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO_2 retention, where a precise inspired FiO_2 is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO_2. The following are estimates of O_2 delivered at the associated flow rates:

LOW-FLOW SYSTEMS

Nasal cannula

Room air 21%(*0.21*)
1L 24%(*0.24*) **4L** 36%(*0.36*)
2L 28%(*0.28*) **5L** 40%(*0.40*)
3L 32%(*0.32*) **6L** 44%(*0.44*)

Simple oxygen masks

- 35-50%(*0.35-0.50*) FiO_2 (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO_2 in the mask

Venturi masks

- 24-50%(*0.24-0.50*) FiO_2 (4-12 L/min)

Partial rebreathing masks

- 40-70%(*0.40-0.70*) FiO_2 (6-10 L/min)

Non-rebreathing masks

- 60-80%(*0.60-0.80*) FiO_2 (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS

Air-entrainment masks/nebulizers

- 24-40%(0.24-0.40) FiO_2

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO_2

31:

High-flow nasal cannula (HFNC) delivers a fraction of inspired oxygen (FiO_2) that is heated (up to 37 degrees Celsius) with 100%(1.0) relative humidity at flow rates up to 60 liters per minute, set independent from the FiO_2. The heated and humidified gas has been shown to improve comfort and aid in clearance of secretions. The high-flow rates reduce anatomical dead space by flushing out carbon dioxide from the upper airways, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less oxygen dilution. A Cochrane review was conducted in 2021 to determine the effectiveness of HFNC compared to standard oxygen therapy and noninvasive positive pressure ventilation (NIPPV) in adult patients requiring respiratory support in the intensive care unit (ICU). The systematic review found that HFNC had less treatment failure compared to standard oxygen therapy (i.e., face mask or nasal cannula with a flow rate of 15 L/min); however, it did not influence mortality rate or length of stay. When HFNC was compared to NIPPV, which included bilevel positive airway pressure (BiPAP) or continuous positive airway pressure (CPAP), there was no difference in treatment failure defined as intubation or re-intubation, mortality, or length of stay. In conclusion, HFNC has been shown to have less treatment failure compared to standard oxygen therapy and may be an alternative to NIPPV in patients treated for acute hypoxic respiratory failure (Lewis et al., Cochrane Database Syst Rev 2021, 3: Cd010172).

32:

Respiratory monitoring includes an evaluation of the patient's respiratory rate, heart rate, oxygen saturation, or work of breathing. Within 90 minutes of high-flow nasal cannula (HFNC) initiation, the patient's respiratory status should improve, indicating adequate respiratory support is being provided. If clinical improvement is not achieved within this time frame, additional respiratory support may be indicated (Lee et al., Intensive Care Med 2013, 39: 247-57).

33:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

34:

Instruction: Labor and delivery criteria may be applied if the delivery of a stillborn or nonviable fetus is warranted.

35:

Instruction: Term labor criteria may be applied for a maximum of two days. If the pregnant individual at term has not delivered after two days of labor, these criteria cannot continue to be utilized.

36:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

37:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

38:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

39:

Relevant complications or comorbidities are conditions that actively require management during an inpatient stay.

40:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

41:

Ultrasonography is utilized to evaluate placental and uterine causes of vaginal bleeding (e.g., retained products of conception within the uterine cavity, placenta accreta, subinvolution of the placental site) (Committee on Practice, Obstet Gynecol 2017, 130: e168-e86).

42:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

43:

A fever is suspected as being new onset when a patient has been afebrile and then develops a new fever or when the fever pattern changes.

44:

There are few studies available to guide the management of hypertension in the postpartum period. Blood pressure typically declines during the first 48 hours post-delivery and then peaks between 3 to 6 days postpartum. A subset of individuals diagnosed with a hypertensive disorder during pregnancy may remain hypertensive after delivery. Prompt treatment with antihypertensive medication is recommended for a systolic blood pressure of 160 mmHg or greater, or a diastolic blood pressure of 110 mmHg or greater which lasts for greater than 15 minutes (Committee on Obstetric, Obstet Gynecol 2017, 129: e90-e5; Powles and Gandhi, CMAJ 2017, 189: E913; American College of Obstetricians and Gynecologists. Obstet Gynecol 2013, 122: 1122-31).

45:

The acute onset of hypertension, with severely elevated systolic blood pressure (BP), diastolic BP, or both, requires immediate treatment with antihypertensive agents for individuals who are pregnant or postpartum. The treatment goal is to decrease BP to a systolic of 140-150 mmHg and a diastolic of 90-100 mmHg. This is done to prevent the loss of cerebral vasculature autoregulation which may be caused by exposure to severe systolic hypertension (Obstet Gynecol 2019, 133: e174-e80).

46:

Initiation of antihypertensive treatment as soon as reasonably possible is recommended for acute-onset severe hypertension (i.e., systolic blood pressure (BP) of 160 mmHg or more or diastolic BP of 110 mmHg or more, or both) that is persistent for 15 minutes or longer(Louis et al., American Journal of Obstetrics and Gynecology 2022, 227: B2-B24; Obstet Gynecol 2020, 135: e237-e60; Obstet Gynecol 2019, 133: e26-e50; Committee on Obstetric, Obstet Gynecol 2017, 129: e90-e5). Drug selection is individualized based on contraindications and risk for adverse outcomes. When BP is under control, oral maintenance medication is initiated to anticipate further treatment needs and prevent a recurrence of severe acute hypertension (Obstet Gynecol 2019, 133: e26-e50). If first-line agents fail to treat severe hypertension effectively, consultation with maternal-fetal medicine, internal medicine, anesthesia, or critical care subspecialists is recommended to discuss second-line agents (Obstet Gynecol 2019, 133: e26-e50; Committee on Obstetric, Obstet Gynecol 2017, 129: e90-e5).

47:

Instruction: It is appropriate to apply this criterion to the patient with stabilized eclampsia after seizure activity has been controlled. A higher level of care is required for the patient who is actively seizing.

48:

Postpartum hypertension (HTN) and preeclampsia refer to the presence of either persistent or exacerbated HTN in individuals who have previously experienced hypertensive disorders of pregnancy or a new-onset development of the condition (Obstet Gynecol 2020, 135: e237-e60). The occurrence of elevated blood pressure during the postpartum period is frequently observed in individuals who had hypertensive disorders of pregnancy during the antepartum period. However, it is also possible for high blood pressure to emerge for the first time following childbirth. Postpartum elevations in blood pressure represent a potential threat to maternal health (Hauspurg and Jeyabalan, Am J Obstet Gynecol 2022, 226: S1211-S21). HTN has emerged as a prominent factor contributing to postpartum readmission in the United States. Furthermore, it is worth noting that more than one-half of maternal deaths occur during the postpartum period. In addition to increased blood pressure, individuals with postpartum preeclampsia frequently present with new-onset headaches or visual disturbances. Timely recognition and intervention are crucial since these individuals are highly susceptible to many medical complications, including stroke, seizures, pulmonary edema, renal failure, congestive heart failure, and mortality (Khedagi and Bello, Cardiol Clin 2021, 39: 77-90). Half of all intracerebral hemorrhage cases attributed to preeclampsia manifest during the postpartum period. A significant number of difficulties related to HTN and preeclampsia manifest during the postpartum period (e.g., 44%(0.44) of eclamptic seizures and 10%(0.10) of maternal deaths) (Asali et al., Pregnancy Hypertens 2019, 17: 133-7).

The existing body of medical literature primarily concentrates on antenatal and intrapartum treatment, with limited available data pertaining to the postpartum period (Hauspurg and Jeyabalan, Am J Obstet Gynecol 2022, 226: S1211-S21; Khedagi and Bello, Cardiol Clin 2021, 39: 77-90; Suresh et al., Obstet Gynecol 2021, 138: 777-87; Asali et al., Pregnancy Hypertens 2019, 17: 133-7). During the initial postpartum period, there are noticeable changes in fluid distribution within the body, which are closely linked to variations in blood pressure. During the first 48 hours following childbirth, there is an initial decline in intravascular fluid levels, followed by a subsequent rise between postpartum days 3 and 6 when the mobilization of fluids occurs. Contributing factors include the administration of intravenous fluids and the loss of pregnancy-associated vasodilation (Khedagi and Bello, Cardiol Clin 2021, 39: 77-90). Several drugs and substances often utilized during the postpartum period have the potential to exacerbate HTN through three primary mechanisms: retention of fluids, activation of sympathomimetic responses, and direct constriction of blood vessels. Of particular significance are nonsteroidal anti-inflammatory medicines (NSAIDs), which are commonly recommended as analgesics in the postpartum period. These pharmaceutical agents reduce the production of prostaglandins, resulting in a diminished capacity for vasodilation and an augmented tendency for sodium retention. It is recommended that NSAIDs be prioritized over opioid analgesics. However, it is essential to consider that individuals with chronic HTN may need increased blood pressure monitoring and modifications to their treatment regimen when using these medications (Obstet Gynecol 2020, 135: e237-e60).

49:

Preeclampsia diagnostic criteria include a systolic blood pressure (BP) of 140 mmHg or higher, or diastolic BP of 90 mmHg or higher, or both, on 2 separate occasions, at least 4 hours apart, after 20 weeks of gestation in a pregnant or postpartum individual with previously normal BP. Proteinuria may be present in preeclampsia. In the absence of proteinuria, new-onset hypertension with new-onset end-organ dysfunction (e.g., impaired liver function) is diagnostic criteria for preeclampsia (Obstet Gynecol 2020, 135: e237-e60). Chronic hypertension with superimposed preeclampsia occurs when preeclampsia is present in a pregnant or postpartum individual with a history of hypertension before pregnancy or 20 weeks of gestation (Obstet Gynecol 2019, 133: e26-e50).

50:

Preeclampsia with severe features is classified by the American College of Obstetricians and Gynecologists (ACOG) as one of the following (Obstet Gynecol 2020, 135: e237-e60):

- Systolic blood pressure (BP) of at least 160 mmHg and/or diastolic BP of at least 110 mmHg on 2 or more measurements at least 4 hours apart (unless antihypertensive therapy is initiated prior to this time, severe hypertension can be confirmed within minutes to facilitate timely intervention)
- Systolic BP of at least 140 mmHg and/or diastolic BP of at least 90 mmHg on 2 or more measurements at least 4 hours apart and one of the following:
    - New-onset headache not related to alternative diagnoses and unresponsive to medication
    - Visual disturbances
    - Pulmonary edema
    - Liver dysfunction indicated by abnormally elevated liver enzymes to more than twice the upper limit of normal or by severe persistent right upper quadrant or epigastric pain unresponsive to medication
    - Renal insufficiency as evidenced by a serum creatinine greater than 1.1 mg/dL or serum creatinine that is doubled in the absence of other renal diseases
    - Thrombocytopenia with platelet count less than 100 x 10^9/L

InterQual® external peer reviewers recommend a higher level of care for individuals who present with pulmonary edema.

51:

New-onset headache associated with preeclampsia is characterized as being persistent, severe, unresponsive to medication (e.g., acetaminophen), and not attributable to an alternative diagnosis (Obstet Gynecol 2020, 135: e237-e60).

52:

Blurred vision and scotomata (e.g., visual field abnormality) are common nervous system visual disturbance manifestations of preeclampsia. In rare circumstances, preeclampsia with severe features can cause temporary vision loss, lasting anywhere from several hours to a week (Obstet Gynecol 2020, 135: e237-e60).

53:

Magnesium sulfate has been shown to reduce the frequency and severity of seizures in pregnant individuals with severe preeclampsia. The rate of seizures in pregnant individuals with severe features of gestational hypertension or preeclampsia is 4 times higher than in those without severe features (e.g., 4 per 200 versus 1 per 200); thus, magnesium sulfate is recommended for the prophylaxis and prevention of seizures in this population (Obstet Gynecol 2020, 135: e237-e60).

54:

Post-dural puncture headache (PDPH), is attributed to the leakage of cerebrospinal fluid caused by a dural puncture occurring after the administration of spinal anesthesia or epidural placement and typically exhibits positional characteristics, such as exacerbation with sitting or standing. PDPH usually manifests within 5 days of a lumbar puncture procedure (Russell et al., Int J Obstet Anesth 2019, 38: 93-103).

55:

The epidural blood patch (EBP) is regarded as the definitive treatment for obstetric post-dural puncture headache (PDHP) (Russell et al., Int J Obstet Anesth 2019, 38: 93-103). EBP should be considered when conservative treatment for obstetric PDHP is ineffective, and the individual has difficulty conducting activities of daily living and caring for the newborn (Russell et al., Int J Obstet Anesth 2019, 38: 104-18).

56:

Conservative management of post-dural puncture headache may include one or more of the following for ≤ 24h:

- Bed rest
- Oral fluids

57:

Most individuals may attain some degree of symptomatic relief of post-dural puncture headache (PDPH) in the supine position; however, this may be transient. The practice of extended bedrest is discouraged due to the heightened susceptibility to thromboembolic complications (Russell et al., Int J Obstet Anesth 2019, 38: 93-103). Dehydration can potentially increase the intensity of PDPH by decreasing cerebrospinal fluid production. Therefore, adequate hydration should be maintained with oral fluids (Uppal et al., Reg Anesth Pain Med 2023:; Russell et al., Int J Obstet Anesth 2019, 38: 93-103).

58:

Pharmacological management of post-dural puncture headache may include two or more of the following:

- Analgesia (includes PO)
- Caffeine (includes PO)
- IV fluids

59:

Dehydration has the potential to enhance the severity of post-dural puncture headache (PDPH) by reducing the production of cerebrospinal fluid. When oral fluid intake is inadequate, intravenous fluid administration is recommended to maintain hydration (Uppal et al., Reg Anesth Pain Med 2023:; Russell et al., Int J Obstet Anesth

2019, 38: 104-18). Caffeine has been recommended for PDPH treatment because it is believed to increase cerebrospinal fluid production by constricting the cerebral vasculature; however, the alleviation of symptoms may be temporary. It is recommended that caffeine therapy be limited to twenty-four hours (Russell et al., Int J Obstet Anesth 2019, 38: 104-18).

60:

Wound dehiscence is associated with a high risk of infection, morbidity, and mortality (Okeahialam et al., Eur J Obstet Gynecol Reprod Biol 2020, 254: 69-73; Villers, Obstet Gynecol Clin North Am 2020, 47: 429-37). Individuals with postpartum wound complications may experience significant wound-healing delays and disruptions. It is essential to note that not all wound complications are associated with infection. Treatment may include re-exploration to evaluate for fascia disruption and necrotizing fascitis, debridement, and reclosure, depending on the severity of the wound and associated complications. Additionally, vacuum-assisted wound closure may be used as an adjunct therapy or alternative to surgery. Education that includes wound care management, methods to prevent surgical site infection, and how to identify and report complications are essential components of care for the patient and their family (Villers, Obstet Gynecol Clin North Am 2020, 47: 429-37).

61:

Hemolysis, elevated liver enzymes, and low platelet count (HELLP) syndrome is a severe form of preeclampsia associated with increased maternal morbidity and mortality. The following criteria are diagnostic benchmarks used to make the diagnosis: lactate dehydrogenase (LDH) 600 IU/L(10.02 µkat/L) or more, aspartate aminotransferase (AST), and alanine aminotransferase (ALT) elevated more than twice the upper limit of normal, and platelet count less than 100,000/cu.mm($100x10^9$/L) (Louis et al., American Journal of Obstetrics and Gynecology 2022, 227: B2-B24; Obstet Gynecol 2020, 135: e237-e60). Due to the serious nature of HELLP syndrome, which is frequently characterized by progressive and sometimes abrupt maternal and fetal deterioration, numerous research articles recommend delivery irrespective of gestational age (Obstet Gynecol 2020, 135: e237-e60). There is a lack of consensus regarding the optimal delivery timing associated with HELLP syndrome when stable maternal and fetal conditions are present (Cavaignac-Vitalis et al., J Matern Fetal Neonatal Med 2019, 32: 1769-75). The management of HELLP syndrome requires very close monitoring, the availability of neonatal and obstetric critical care, and personnel with specialized expertise; therefore, pregnant individuals who are far removed from term should be treated at a tertiary care facility. Various hematologic and hepatic changes can occur in pregnant individuals with severe preeclampsia or HELLP syndrome. Platelet activation, aggregation, and consumption can lead to thrombocytopenia and represent a disease severity marker. Elevated serum concentrations of LDH can be a sign of hemolysis, and elevated LDH, AST, and ALT serum levels indicate hepatic dysfunction. Evaluation of coagulation parameters is useful when the platelet count is below 150,000/cu.mm($150x10^9$/L), with significant liver dysfunction or suspected placental abruption. Abnormalities of prothrombin time, partial prothrombin time, and fibrinogen are reflective of hepatic synthetic dysfunction and advanced preeclampsia (Obstet Gynecol 2020, 135: e237-e60).

62:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

63:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

64:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

65:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

66:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

67:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

68:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours

- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

69:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

70:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

71:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.
Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

72:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

73:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

74:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

75:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

76:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

77:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.
The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some

physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

78:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

79:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

80:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

81:

Notable causes for prolonged mechanical ventilation include:
- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):
- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

82:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.